Low-Level Vaccination to Prevent Disease Emergence

Can we stop vaccinating against pox viruses?

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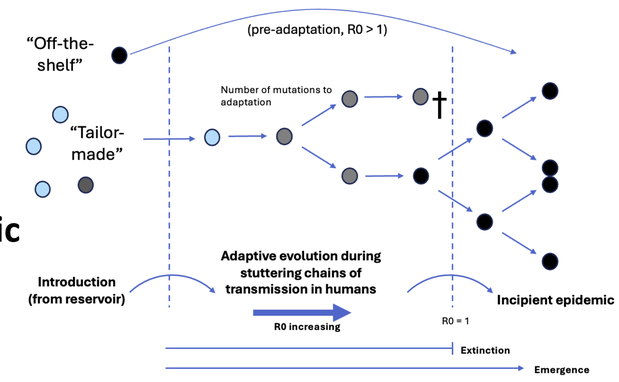
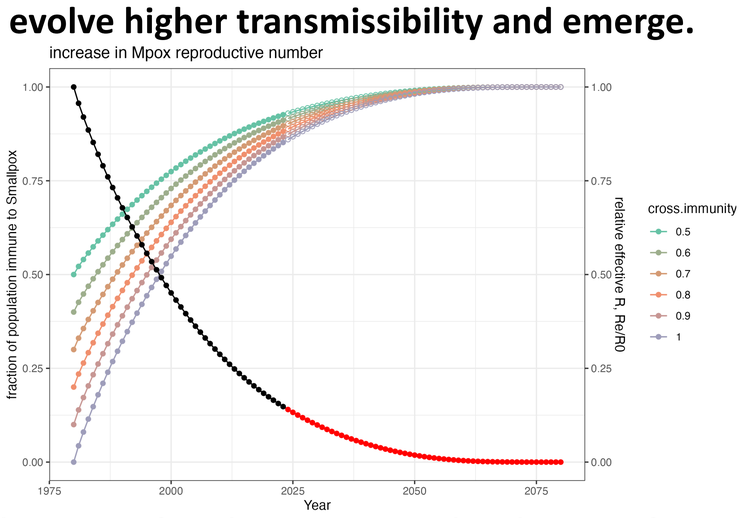
Motivation

Zoonotic pathogens can acquire R > 1 in a new population in two ways: 1) accumulating the necessary mutations while circulating in the reservoir ("off the shelf"), or **2) accumulating the necessary mutations during stochastic transmission *after spillover* ("tailor made").**

After the eradication of a disease with no known zoonotic reservoir, **population immunity due to natural infection and vaccination wanes.**

This provides an opportunity for related zoonotic viruses with limited human-human spread to evolve higher transmissibility and emerge.





(Projections of population immunity and R_e relative to R_0 for Mpox in the DRC)

Because Smallpox has no known zoonotic reservoir, vaccination could end after eradication.

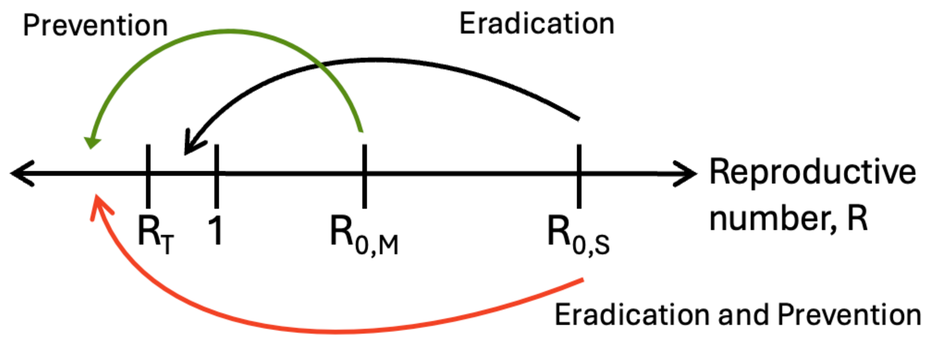
As herd immunity wanes, the reproductive numbers (R_es) of related zoonotic orthopoxviruses, including Mpox, are likely increasing, potentiating selection for higher transmissibility during longer chains of transmission.

Big Idea and Takeaways

We developed a multi-type branching process model to explore the evolutionary potential of pathogens with reproductive numbers R < 1 to evolve sustained transmission (R > 1) during stochastic chains of transmission after spillover into a new host population.

We use this model to investigate the potential of vaccination as a strategy to prevent the emergence of highly transmissible zoonotic disease.

$$F_p = 1 - \frac{R_T}{R_0}$$



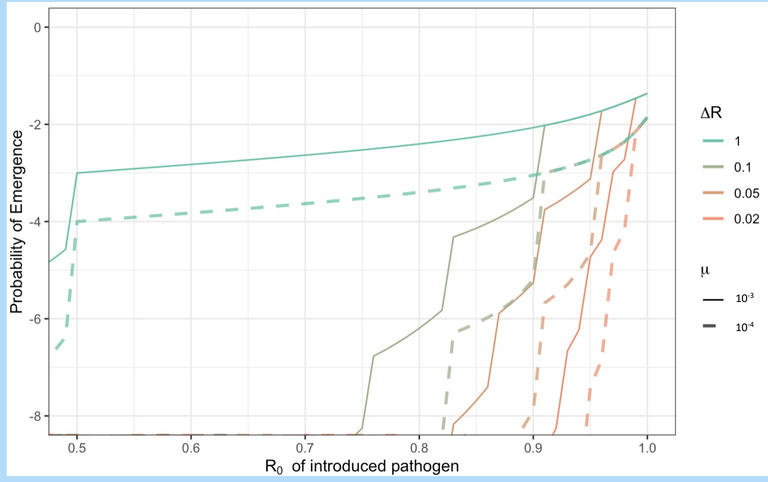
The fraction of a population which must be vaccinated to prevent the adaptive evolution of a zoonotic pathogen, $\mathbf{F_p}$, depends on the basic reproductive number, $\mathbf{R_0}$, and the threshold reproductive number, $\mathbf{R_T}$, above which evolution during transmission is likely.

We find that:

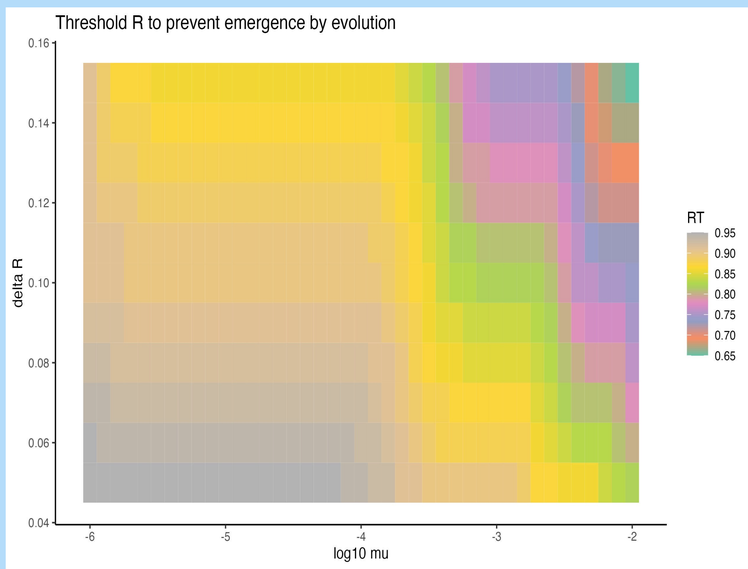
1) **Pathogens with R < 1 can evolve sustained human-human transmission (R > 1) during outbreaks;**
2) **The probability of emergence of a pathogen with R > 1 increases sharply with the reproductive number of the spillover infection; and**
3) **Modest levels of vaccination can dramatically reduce the probability of disease emergence by adaptive evolution in transmission chains.**

Results and Conclusions

Probability of evolving sustained transmission (R > 1) increases sharply with RE of the spillover infection (R_e < 1).

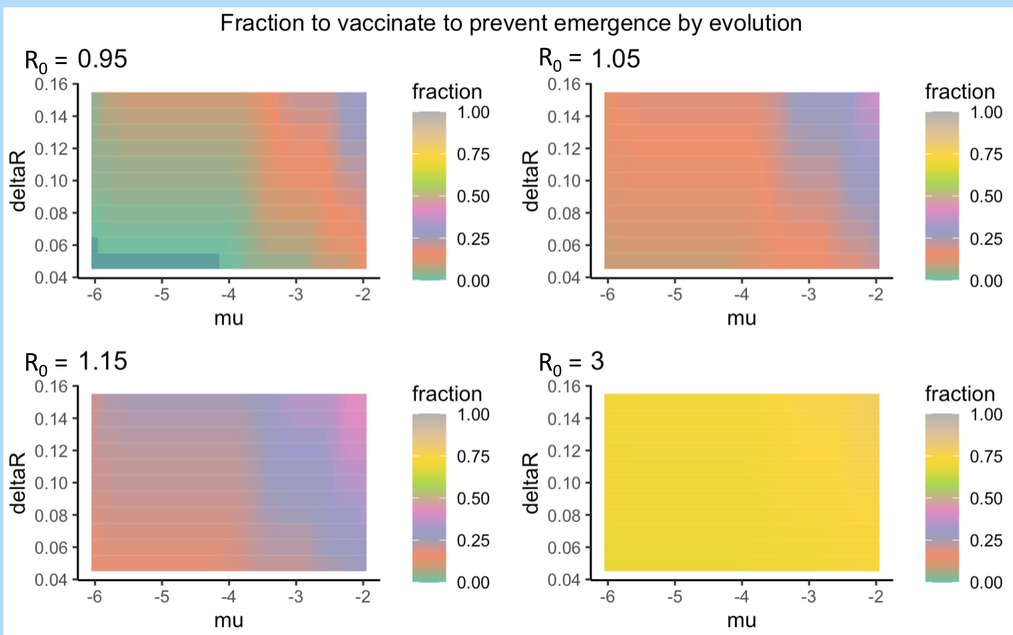


As R_e increases, fewer mutations are required to evolve R > 1, introducing steep step-like increases in emergence probability. At $P_E(R_T) = 10^{-5}$, 1 in 100,000 spillovers on average evolves to sustained transmission. Delta R: proportional increase in transmissibility with mutation; mu: rate of adaptation.



There exists some R_e < 1, R_T, above which evolution of sustained transmission during an outbreak may occur in the absence of intervention.

Relatively little vaccination can dramatically reduce the probability of evolving sustained transmission for zoonoses with limited human-human spread.



Reducing the probability of emergence below 1 in 10^5 spillovers requires significantly less vaccination than eradicating and preventing re-emergence after the pathogen evolves a high R.

Preventing the emergence of zoonotic disease requires constant vaccination to maintain herd preventative immunity. This involves **significantly less vaccination** than eradication once endemic.